Reproducing the Shiny app workflow

This vignette reproduces the main path through the Spatial Methods Workbench using small, deterministic simulated data. The same seed produces the same CSV and statistical output. No protected data, private method, or unpublished manuscript content is required.

1. Create upload-ready demonstration files

```
spatial <- demo_spatial_data(n = 72, seed = 2026)
image <- demo_image_data(n = 48, features = 8, seed = 2026)
contour <- demo_contour_data(contours = 6, points = 16, seed = 2026)
volume <- demo_spatial_3d_data(n = 72, volumes = 2, seed = 2026)

vapply(list(spatial = spatial, image = image, contour = contour, volume = volume), nrow, integer(1))
#> spatial   image contour  volume
#>      72      48      96      72
```

Write any dataset to CSV, then upload it through **01 · DATA** in the app.

```
demo_dir <- file.path(tempdir(), "spatial-workbench-vignettes")
dir.create(demo_dir, showWarnings = FALSE)
utils::write.csv(spatial, file.path(demo_dir, "spatial_demo.csv"), row.names = FALSE)
utils::write.csv(image, file.path(demo_dir, "image_demo.csv"), row.names = FALSE)
utils::write.csv(contour, file.path(demo_dir, "contour_demo.csv"), row.names = FALSE)
utils::write.csv(volume, file.path(demo_dir, "spatial_3d_demo.csv"), row.names = FALSE)
basename(list.files(demo_dir))
#> [1] "contour_demo.csv"    "image_demo.csv"      "spatial_3d_demo.csv"
#> [4] "spatial_demo.csv"
```

Start the app from the repository root with `shiny::runApp()` or use the live application.

2. Reproduce column mapping

For `spatial_demo.csv`, select the following roles in **02 · COLUMN MAP**.

App role	Column
X coordinate	`x`
Y coordinate	`y`
Region / zone	`zone`
Sample / contour / volume ID	`sample_id`
Outcome	`disease`
Exposure	`epithelial`
Mediator	`APC`
Numeric features	`epithelial`, `APC`, `CTL`, `Treg`, `T_helper`

The programmatic equivalent is:

```r
mapping <- coerce_mapping(spatial, list(
  x = "x", y = "y", z = NULL, zone = "zone", id = "sample_id",
  outcome = "disease", exposure = "epithelial", mediator = "APC"
))
features <- c("epithelial", "APC", "CTL", "Treg", "T_helper")
validate_dataset(spatial, mapping)[c("ok", "n", "p", "missing_rate")]
#> $ok
#> [1] TRUE
#>
#> $n
#> [1] 72
#>
#> $p
#> [1] 10
#>
#> $missing_rate
#> [1] 0
```

3. Reproduce readiness checks

```r
readiness <- recommend_methods(spatial, mapping, features)
data.frame(
  analysis = names(readiness),
  compatible = vapply(readiness, `[[`, logical(1), "compatible"),
  reason = vapply(readiness, `[[`, character(1), "reason"),
  row.names = NULL
)
#>               analysis compatible
#> 1           spatial_qc       TRUE
#> 2  conditional_network       TRUE
#> 3                ispat       TRUE
#> 4                gpghs       TRUE
#> 5              ispat3d      FALSE
#> 6           spatial_ml       TRUE
#> 7   spatial_clustering       TRUE
#> 8    neural_prediction       TRUE
#> 9         scalar_image      FALSE
#> 10      image_to_image      FALSE
#> 11           mediation      FALSE
#> 12           shape_pca      FALSE
#>                                                                  reason
#> 1
#> 2
#> 3
#> 4
#> 5                                                             Select z
#> 6
#> 7
#> 8
#> 9                   Select a numeric outcome for wide-image regression
#> 10      Select at least three input__ and three output__ image columns
#> 11                 Exposure, mediator, and outcome must all be numeric
#> 12 Select an ID with at least five contours and six points per contour
```

Readiness describes whether the mapped table satisfies each input contract and whether an optional engine is installed. It is not a scientific recommendation.

4. Reproduce an app run

In **03 · ANALYSIS**, choose **Spatial exploration & autocorrelation**, select `epithelial`, set six neighbors and 19 permutations, acknowledge the citation condition, and run. The equivalent call is:

```
result <- run_analysis(
  "spatial_qc", spatial, mapping, features,
  params = list(feature = "epithelial", neighbors = 6, permutations = 19, seed = 2026)
)
result$table
#>                   metric      value
#> 1 Complete observations 72.0000000
#> 2             Neighbors  6.0000000
#> 3             Moran's I  0.7019779
#> 4   Permutation p-value  0.0500000
```

```
result$plot
```

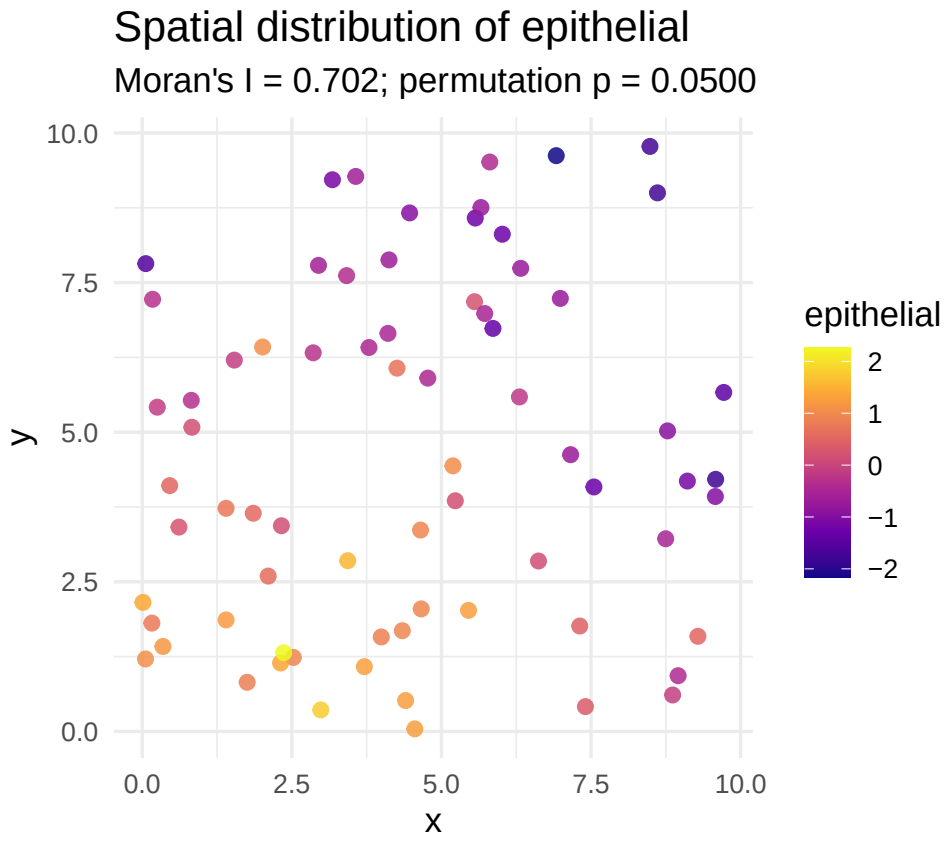

Figure 1: Synthetic-data output from the same workflow used by the app.

The table, plot, summary, controls, mappings, citations, and fitted result object are retained. This vignette does not interpret the numerical result.

5. Reproduce the download

```
bundle <- file.path(tempdir(), "spatial-workbench-example.zip")
write_result_bundle(result, bundle)
utils::unzip(bundle, list = TRUE)$Name
#> [1] "CITATION.txt"      "COLLABORATION.txt" "figure.png"
#> [4] "FUNDING.txt"       "manifest.json"     "result_object.rds"
#> [7] "results.csv"
```

The archive contains the numerical table, figure, manifest, R object, citations, and collaboration notice. It does not contain the uploaded data or an automated interpretation.